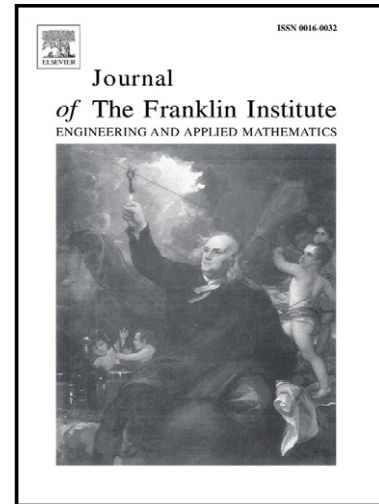


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Fuzzy Control for Electric Power Steering System with Assist

Motor current input constraints

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Abstract – Friction and disturbances of the road are the main sources of nonlinearity in the Electric Power Steering (EPS) system. Consequently, conventional linear controllers design based on a simplified linear model of the EPS system will result in poor dynamic performance or system instability. On the other hand, a brush-type DC motor is more used in EPS control with an input current that is limited in practice. The control laws designed without taking into account the saturation effect may have undesirable consequences on the system stability. In this paper, a Takagi-Sugeno (T-S) fuzzy is used to represent the nonlinear behavior of an EPS system, and stabilization conditions for nonlinear EPS system with both constrained and saturated control input cases are proposed in terms of linear matrix inequalities (LMI). Simulation results show that both the saturated and constrained controls can stabilize the resulting closed-loop EPS system and provide a stable driving in the presence of nonlinear friction, disturbance of the road and actuator saturation.

Keywords: Electric Power Steering (EPS), Actuator saturation, Takagi-Sugeno (T-S) fuzzy systems, Linear Matrix Inequality (LMI).

1. Introduction

In recent years, the conventional hydraulic steering system has been replaced by Electric Power Steering (EPS) System, because its modularity, tenability of steering feel, and environmental friendliness [1].

The main roles of EPS systems consist in improving steering feels levels as perceived by the driver. To achieve this goal, the conventional linear controls based on simplified linear models of the EPS system are applied in [1-3]. However, the EPS system is a highly nonlinear system because of the nonlinear friction and the disturbance of the road [4]. Consequently, the conventional linear controls will result in poor dynamic performance or system instability. In order to resolve this problem, recent works, such as, a sliding mode controller [5], fuzzy controller [6] and an H_∞ control [7] are applied to EPS systems. Nevertheless, in the aforementioned papers, they have not taken into account the nonlinearity effect of actuator saturation. Indeed, a brush-type DC motor is more used in EPS control with an input current that is limited in practice. However, the constraints on the control input may degrade the performance of the closed-loop system and can lead to instability for large disturbances [8-17]. Many results are reported in the literature regarding stability analysis of linear systems (more recently, see [8-10]) and nonlinear systems [11-17] subject to input constraints. In most cases, to deal with this problem, the bound on the initial condition is given to avoid the input saturation in low-gain control laws for nonlinear systems [11, 12, 14, 15]. Otherwise, the domain of attraction is estimated in the presence of actuator saturation [11, 12, 14-18]. More specifically, the saturation is transformed into a dead-zone nonlinearity in [18], and it is represented by a polytopic model in [11, 12, 14-17].

In this paper, a T-S fuzzy method is used to represent the nonlinear behavior of an EPS system, and stabilization conditions for the nonlinear EPS system with both constrained and saturated control input cases are proposed in terms of linear matrix inequalities (LMIs).

This paper is organized as follows. Section 2 gives a mathematical description of the EPS model. Section 3 gives a Takagi-Sugeno fuzzy model of the EPS system. In Section 4, a controller with actuators saturation based on the T-S fuzzy model is designed. In Section 5, simulation results are given to show the effectiveness of the proposed method. Finally, the paper is concluded in Section 6.

Notation: $\Re$ denotes the set of real numbers and $\Re^{n \times m}$ is the set of all $n \times m$ real matrices. $M > (\geq, <, \leq) 0$ is used to denote a symmetric positive definite (positive semi-definite, negative definite, negative semi-definite, respectively) matrix. $*$ denotes the symmetric block matrix, $X + (*)$ denotes $X + X^T$, $\times$ denotes the multiplication and co denotes the convex hull.

2. Nonlinear model of EPS system

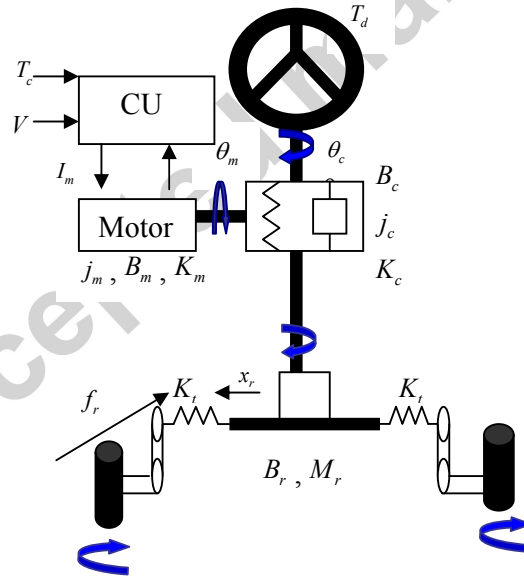


Figure1. EPS system

The EPS system has three basic components (see Figure 1): the control unit (CU) consisting of all sensors and controllers, an assist motor and mechanical steering such as steering wheel, steering column and steering rack. Its basic operating principle can be summarized as follows: When the driver turns the steering wheel, the steering rack will move the vehicle wheels in

the appropriate direction and a rack force $f_r(t)$ is generated. In order to compensate for this force, the motor applies the servo force $T_m(t)$. The role of CU is to determine the motor current I_m in order to obtain a good steering feeling.

In Figure 1, $\theta_c(t)$ is the steering hand wheel angle, T_c is the steering torque, $\theta_m(t)$ is the angular position of the motor, x_r is the rack position, $f_r(t)$ is the rack force, I_m is the input current of the motor, j_c is the steering column moment of inertia, B_c is the steering column viscous damping, j_m is the motor moment of inertia, K_m is the motor torsional stiffness, B_m is the motor damping, M_r is Rack and wheel assembly mass, B_r is damping of the rack, and K_t is the tire spring rate.

The friction and disturbances of the road are the main sources of nonlinearity in the EPS system. Consequently, conventional linear controls based on a simplified linear model of the EPS system will result in poor dynamic performance or system instability. In this paper, the proposed model is obtained by considering the aforementioned nonlinearities.

The dynamic model of the EPS system is given by the relation between mechanical steering system, a brush-type DC motor and tire/road contact forces as follows [6]:

$$\begin{cases} \dot{\theta}_c(t) = w_c(t) \\ \dot{w}_c(t) = \frac{1}{j_c} (T_d - T_c - B_c w_c - F_c \text{sign}(w_c(t))) \\ \dot{\theta}_m(t) = w_m(t) \\ \dot{w}_m(t) = \frac{1}{j_m} (T_m(t) - T_d(t) - B_m w_m - F_m \text{sign}(w_m(t))) \\ \dot{x}_r(t) = v_r(t) \\ \dot{v}_r(t) = \frac{1}{M_r} \left(\frac{T_c + G_G T_d(t)}{r_p} - K_t x_r - f_r(t) - B_r v_r - F_r \text{sign}(v_r(t)) \right) \end{cases} \quad (1)$$

where r_p is the radius of the pinion, M_r is the rack and wheel assembly mass, G_G is the motor gear ratio, F_c is the steering column friction, F_m is the motor friction, F_r is the rack motor friction, and the effective assisting torque $T_a(t)$ is given by:

$$T_a(t) = K_a I_m \quad (2)$$

where K_a is the motor torque constant. The steering torque force T_c is given by:

$$T_c = K_c \left(\theta_c(t) - \frac{x_r(t)}{r_p} \right) \quad (3)$$

and the servo force $T_m(t)$ is expressed by :

$$T_m(t) = K_m \left(\theta_m(t) - \frac{G_G x_r(t)}{r_p} \right) \quad (4)$$

2.1 Vehicle model

The rack force $f_r(t)$ is generated by the road reaction force as shown in block diagram of Figure 2. This force can be derived using the vehicle model with single track, so-called bicycle model.

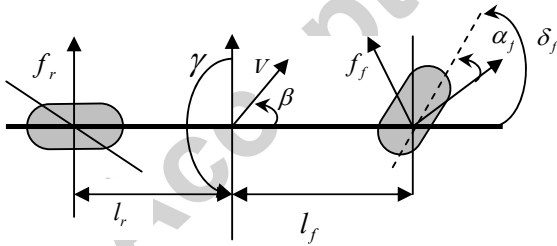


Figure 3. vehicle model with single track

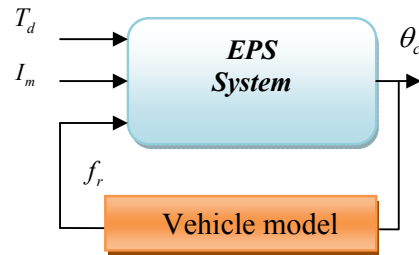


Figure 2. System block diagram

Figure 3 shows the model of the single-track vehicle where β denotes the side slip angle and γ is the yaw rate. The following vehicle lateral dynamics are obtained by considering small

angle approximations, lateral tire force is assumed to be proportional to the tire slip angle and the vehicle is assumed running at a constant speed V .

$$\begin{cases} \dot{\beta}(t) = \frac{1}{MV} \left[-(C_f + C_r) \beta(t) - \left(MV - \frac{1}{V} (l_f C_f - l_r C_r) \right) \gamma(t) + C_f \delta_f \right] \\ \dot{\gamma}(t) = \frac{1}{I_z} \left[-(l_f C_f - l_r C_r) \beta(t) - \frac{1}{V} (l_f^2 C_f + l_r^2 C_r) \gamma(t) + l_f C_f \delta_f \right] \end{cases} \quad (5)$$

where l_f is the chassis length of front, l_r is the chassis length rear, C_f is the cornering stiffness coefficient, C_r is the cornering stiffness coefficient and I_z is the moment of vehicle inertia.

The rack force $f_r(t)$ is described by the following expression:

$$f_r(t) = \frac{T_p C_f \left[\delta_f - \left(\beta(t) + \frac{l_f}{V} \gamma(t) \right) \right]}{r_p} \quad (6)$$

In expression (6), T_p is the caster trail and the front steer angle δ_f is given by:

$$\delta_f = \frac{\theta_c}{G_{sc}} \quad (7)$$

where G_{sc} is the steering system ratio.

3. Takagi-Sugeno representation of EPS systems

The Takagi-Sugeno fuzzy systems provide an effective representation of EPS systems by a set of linear models. This approach has been extensively used for stability analysis and control of nonlinear systems (see for example [11-23] and the references therein). A T-S fuzzy system with external disturbances and actuators saturation is represented by the following fuzzy rules [15]:

R_i : If $\xi_I(t)$ is about M_{I_i} and... and $\xi_q(t)$ is about M_{q_i}

$$\text{THEN } \begin{cases} \dot{x}(t) = A_i x(t) + B_{1i} w(t) + B_{2i} \sigma(t) \\ y(t) = C_{2i} x(t) \end{cases}, i \in I_r \quad (8)$$

in which M_{ji} is the fuzzy set of ξ_i in rule R_i , r is the number of IF-THEN fuzzy rules and $\xi(t) \in \mathfrak{R}^q$ is the decision variable vector and $\xi_1(t), \xi_2(t), \dots, \xi_q(t)$ are the premise variables assumed measurable, $x(t) \in \mathfrak{R}^n$ is the system state vector, $\sigma(t) \in \mathfrak{R}^m$ is the saturated control input, $y(t) \in \mathfrak{R}^p$ is the measurable output and $w(t)$ is the disturbance variable.

The global dynamic system is inferred as:

$$\begin{cases} \dot{x}(t) = \sum_{i=1}^r \mu_i(\xi(t)) (A_i x(t) + B_{1i} w(t) + B_{2i} \sigma(t)) \\ y(t) = \sum_{i=1}^r \mu_i(\xi(t)) C_{2i} x(t) \end{cases} \quad (9)$$

where

$$\mu_i(\xi(t)) = \frac{\prod_{j=1}^q M_{ji}(\xi_j(t))}{\sum_{i=1}^r \left(\prod_{j=1}^q M_{ji}(\xi_j(t)) \right)} \quad (10)$$

$M_{ji}(\xi_j(t))$ is the grade of membership of $\xi_j(t)$ in M_{ji} . The normalized activation function $\mu_i(\xi(t))$ in relation with the i^{th} sub-model is such that:

$$\begin{cases} \sum_{i=1}^r \mu_i(\xi(t)) = 1 \\ 0 \leq \mu_i(\xi(t)) \leq 1 \quad \forall i \in I_r \end{cases} \quad (11)$$

Now, assumed that; $w_{cmin} \leq w_c(t) \leq w_{cmax}$, and consider $w_c(t)$, $w_m(t)$ and $v_r(t)$ have the same sign, the Takagi-Sugeno representation of the EPS system can be written as:

$$\begin{aligned} \dot{x}(t) &= \sum_{i=1}^r \mu_i(w_c(t)) (A_i x(t) + B_{1i} w(t) + B_{2i} \sigma(t)) \\ y(t) &= Cx(t) \end{aligned} \quad (12)$$

where

$$\begin{aligned} x(t) &= [\theta_c \quad w_c \quad \theta_m \quad w_m \quad x_r \quad v_r \quad \beta \quad \gamma]^T, y(t) = T_c, w(t) = T_d, \\ \mu_1(t) &= \frac{w_c - w_{cmin}}{w_{cmax} - w_{cmin}}, \mu_2(t) = \frac{w_{cmax} - w_c}{w_{cmax} - w_{cmin}}, r = 2. \end{aligned}$$

$$\begin{aligned}
 A_1 &= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ a_{21} & a_{22}^1 & 0 & 0 & a_{51} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & a_{43} & a_{44}^1 & 0 & a_{46} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ a_{61} & 0 & a_{63} & 0 & a_{65} & a_{66}^1 & a_{67} & a_{68} \\ a_{71} & 0 & 0 & 0 & 0 & 0 & a_{77} & a_{78} \\ a_{81} & 0 & 0 & 0 & 0 & 0 & a_{87} & a_{88} \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ a_{21} & a_{22}^2 & 0 & 0 & a_{51} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & a_{43} & a_{44}^2 & 0 & a_{46} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ a_{61} & 0 & a_{63} & 0 & a_{65} & a_{66}^2 & a_{67} & a_{68} \\ a_{71} & 0 & 0 & 0 & 0 & 0 & a_{77} & a_{78} \\ a_{81} & 0 & 0 & 0 & 0 & 0 & a_{87} & a_{88} \end{bmatrix} \\
 B_1 &= \begin{bmatrix} 0 & \frac{1}{j_c} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T, B_2 = \begin{bmatrix} 0 & 0 & 0 & \frac{K_a}{j_m} & 0 & 0 & 0 & 0 \end{bmatrix}^T \\
 C &= \begin{bmatrix} K_c & 0 & 0 & 0 & \frac{-K_c}{r_p} & 0 & 0 & 0 \end{bmatrix} \\
 a_{21} &= \frac{-K_c}{j_c}, a_{22}^1 = \frac{-B_c - F_c}{j_c}, a_{25} = \frac{K_c}{j_c \cdot r_p}, a_{43} = \frac{-K_m}{j_m}, a_{44}^1 = \frac{-B_m - F_m}{j_m}, a_{46} = \frac{G_g \cdot K_m}{j_m \cdot r_p} \\
 a_{61} &= \frac{K_c}{M_r \cdot r_p} - \frac{T_p \cdot C_f}{G_{sc} \cdot M_r \cdot r_p}, a_{63} = \frac{G_g \cdot K_m}{M_r \cdot r_p}, a_{65} = -\frac{K_c + K_t \cdot r_p^2 + G_g^2 \cdot K_m}{M_r \cdot r_p^2}, a_{66}^1 = \frac{-B_r - F_r}{M_r} \\
 a_{67} &= \frac{T_p \cdot C_f}{M_r \cdot r_p}, a_{68} = \frac{T_p \cdot C_f \cdot l_f}{V \cdot M_r \cdot r_p}, a_{71} = \frac{C_f}{G_{sc}}, a_{77} = -\frac{C_f + C_r}{V \cdot M}, a_{78} = \frac{C_f \cdot l_f - C_r \cdot l_r - V^2 \cdot M}{V^2 \cdot M} \\
 a_{81} &= \frac{C_f \cdot l_f}{I_z \cdot G_{sc}}, a_{87} = -\frac{C_f \cdot l_f - C_r \cdot l_r}{I_z}, a_{88} = -\frac{C_f \cdot l_f^2 + C_r \cdot l_r^2}{I_z \cdot V}, a_{22}^2 = \frac{-B_c + F_c}{j_c}, a_{44}^2 = \frac{-B_m + F_m}{j_m} \\
 a_{66}^2 &= \frac{-B_r + F_r}{M_r}
 \end{aligned}$$

4. Fuzzy control design with actuator saturation

A brush-type DC motor is widely used in EPS controller. The battery pack is the heart of an electric vehicle and its power is limited [24]. This means that, a brush-type DC is subject to input constraint as:

$$-\bar{I} \leq I_m \leq \bar{I} \quad (13)$$

where $\bar{I}$ denotes the saturation level (see Figure 4). The control laws designed without taking into account the saturation effect may have undesirable consequences on the system. The saturation function is defined as follows:

$$\sigma(t) = \text{sat}(I_m) = \text{sign}(I_m) \min\{I_m, |I_m|\} \quad (14)$$

Let the set $\mathfrak{I}(H_j)$ be defined as:

$$\mathfrak{I}(H_j) = \{x(t) \in R^n \mid |h_i^j x| \leq \bar{u}_i\} \quad (15)$$

where H_j is an $m \times n$ matrix and h_i^j is the i^{th} row of the matrix H_j , m is the number of control inputs and n is the number of state variables (in our application $m=1, n=8$).

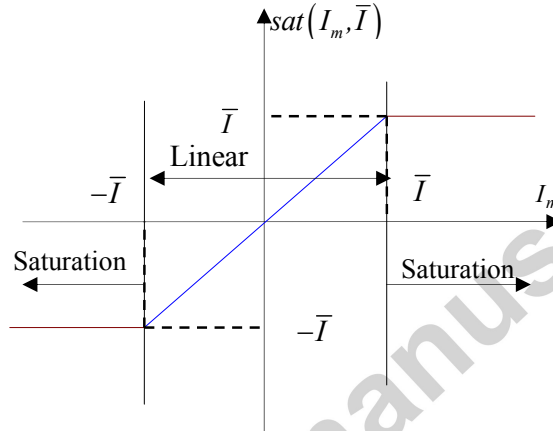


Figure 4. Saturation function

For all $x(t) \in \mathfrak{I}(H_j)$, saturation effect (14) can be written in the following polytopic representation [25].

$$\begin{cases} \text{sat}(I_m(t), \bar{I}) = \sum_{s=1}^{2^m} \alpha_s (E_s I_m(t) + E_s^{-1} \nu) \\ \sum_{i=1}^{2^m} \alpha_s = 1, \quad 0 \leq \alpha_s \leq 1 \\ \nu(t) = \sum_{j=1}^r \mu_j(\xi(t)) H_j x(t) \end{cases} \quad (16)$$

E_s denotes all elements of E , and $E_s^{-1} = I - E_s$,

The controller has the following form:

$$I_m(t) = \sum_{j=1}^r \mu_j(\xi(t)) K_j x(t) \quad (17)$$

where $K_j \in R^{m \times n}$ is the local controller matrix to be determined. The control strategy is shown

in Figure 5.

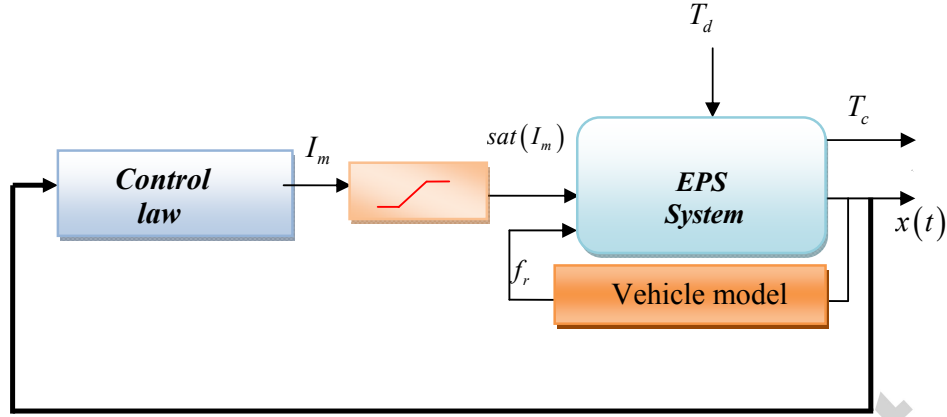


Figure 5. Closed-loop system block diagram

4.1 Constrained control

In this control, we will design the control input (17) in such away to avoid saturation limits (the current cannot be saturated). Therefore, the saturation function (14) becomes linear and one has:

$$\sigma(t) = I_m(t) = \sum_{j=1}^r \mu_j(\xi(t)) K_j x(t) \text{ for all } x(t) \in \bigcap_{j=1}^r \mathfrak{Z}(K_j) \quad (18)$$

with $\mathfrak{Z}(K_j) = \{x(t) \in R^n \mid |K_j x| \leq \bar{I}\}$ for all $j \in \{1, 2\}$.

The closed-loop system composed by (12) and (18) is described by:

$$\begin{cases} \dot{x}(t) = \left(\sum_{i=1}^r \mu_i(\xi(t)) A_i + B_2 \sum_{j=1}^r \mu_j(\xi(t)) K_j \right) x(t) + B_1 w(t) \\ y(t) = Cx(t) \end{cases} \quad (19)$$

For a symmetric positive definite matrix $P \in R^{n \times n}$, we define the Lyapunov function as follows:

$$V(x) = x(t)^T P x(t) \quad (20)$$

An ellipsoid $\varepsilon(P) = \{x \in R^n \mid x^T P x \leq 1\}$ is said to be contractively invariant set if $\dot{V}(x) < 0$,

$\forall x \in \varepsilon(P) \setminus \{0\}$. Therefore, if an ellipsoid is contractively invariant, it is inside the domain of attraction [11].

The following lemma is practically a special case of the theorem proposed in [11] and will be used for the control law design.

Lemma 1: For a given system (12) and a given state-feedback control (18), the ellipsoid $\varepsilon(P, \rho)$ is contractively invariant set of the closed-loop system (19), if there exist a symmetric positive definite matrix Q and matrices $N_j \in R^{m \times n}$ as a solution to the following LMIs:

$$\begin{aligned} & i. A_i Q + B N_i + (*) < 0, i \in I_r \\ & ii. A_i Q + B N_j + A_j Q + B N_i + (*) \leq 0, i < j \leq r \\ & iii. \begin{bmatrix} 1 & N_j \\ * & Q \end{bmatrix} \geq 0, j \in I_r \end{aligned} \quad (21)$$

Then, the control law (18) is given by $K_j = N_j Q^{-1}$.

Proof: see [11][17].

4.2 Saturated control design

In this control design, our goal is to design a nonlinear controller in the presence of saturation.

Thus, we have:

$$\sigma(t) = \text{sat}(I_m, \bar{I}) \quad (22)$$

Using the definition (15), the closed-loop system composed by (12) and (18) is described by:

$$\begin{cases} \dot{x}(t) = \sum_{i=1}^r \mu_i(\xi(t)) \sum_{j=1}^r \mu_j(\xi(t)) \sum_{s=1}^{2^m} \alpha_s (A_i + B_2 (E_s K_j + E_s H_j)) x(t) + B_1 w(t) \\ y(t) = Cx(t) \end{cases} \quad (23)$$

The stabilization conditions of closed-loop system (23) can be obtained by the following lemma:

Lemma 2: For a given system (12) and a given state-feedback control (18), the ellipsoid $\varepsilon(P, \rho)$ is contractively invariant set of the closed-loop system (23) if there exist a symmetric positive definite matrix Q and matrices $N_j \in R^{m \times n}, Z_j \in R^{m \times n}$ as a solution to the following LMI problem:

$$\begin{aligned}
& i. \left(A_i Q + B_2 (E_s N_i + \bar{E}_s Z_i) \right) + (*) < 0, s \in I_{2^m} < 0, i \in I_r \\
& ii. \left[\left(A_i Q + B_2 (E_s N_j + \bar{E}_s Z_j) \right) + \left(A_j Q + B_2 (E_s N_i + \bar{E}_s Z_i) \right) \right] + (*) \leq 0, 1 \leq i < j \leq r, s \in I_{2^m} \quad (24) \\
& iii. \begin{bmatrix} 1 & * \\ (z_i^j)^T & Q \end{bmatrix} \geq 0, i \in I_m, j \in I_r
\end{aligned}$$

Then, the control law (18) is given by $K_j = N_j Q^{-1}$ with the matrix $H_j = Z_j Q^{-1}$.

Proof: see [11][17].

5. Simulation results

Table 1. values of EPS model parameters

<i>Symbol</i>	<i>values</i>
j_c	11.4 Kg.m ²
B_c	0.072 N.m./(rad/s)
K_c	114.6 N.m /rad.
F_c	0.0052 N.m
j_m	5.10 ⁻⁴ Kg.m ²
B_m	0.032 N.m./(rad/s)
K_m	125 N.m
K_a	0.05 N.m/A
F_m	0.0052 N.m
M	1814 kg
M_r	32 kg
F_r	0.002 N.m
K_t	32900 N.m
l_r	1.8m
l_f	1m
I_z	4240 Kg.m ²
r_p	0.007m
V	20m/S
C_f	126000 N/rad
C_r	126000 N/rad
T_p	0.033m
G_{Gs}	20
G_G	16.5

[illegible]

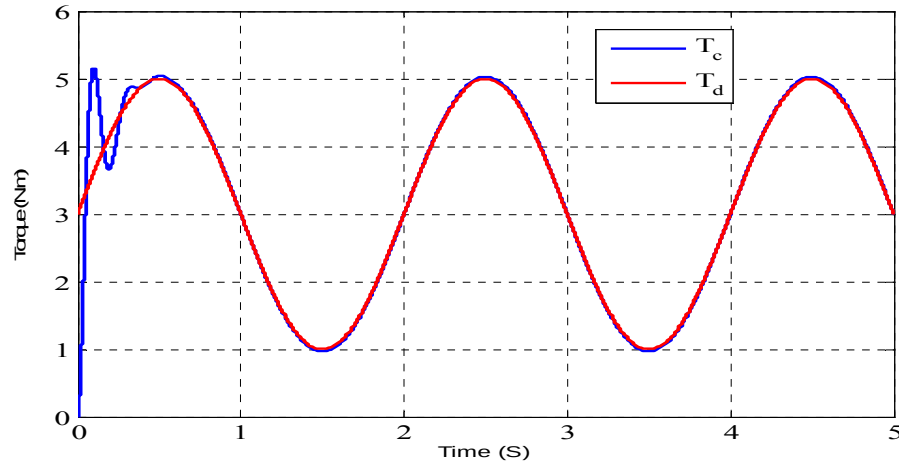


Figure 6. The trajectory of torque T_c to T_d in saturated control at $0.5Hz$

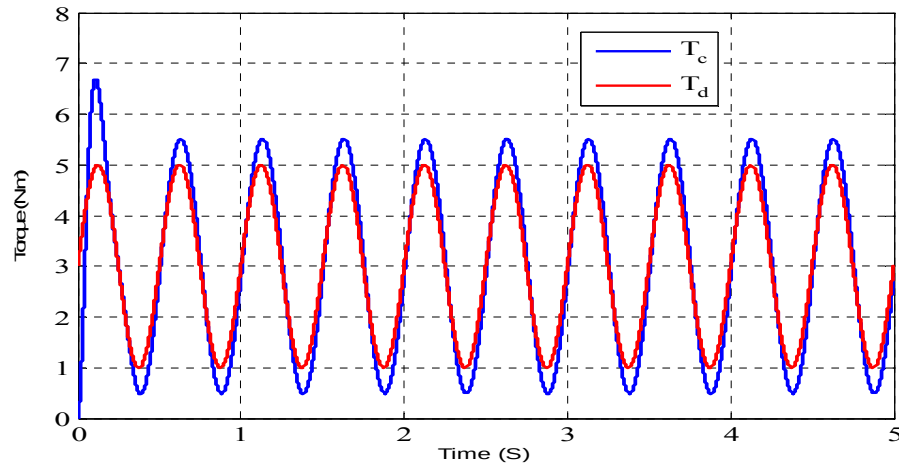


Figure 7. The trajectory of torque T_c to T_d in saturated control at $2Hz$

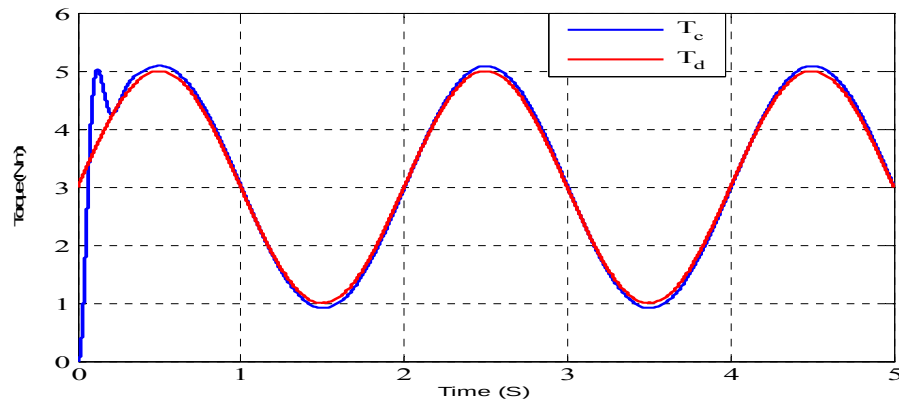


Figure 8. The trajectory of torque T_c to T_d in constrained control at $0.5Hz$

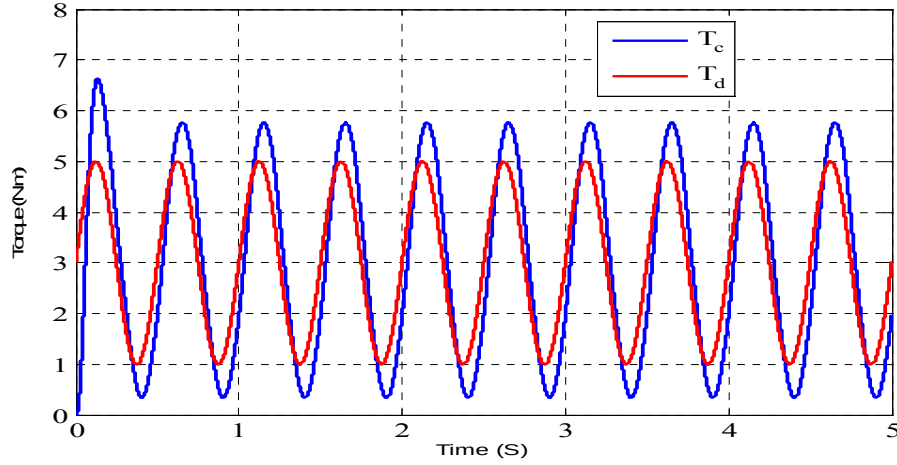


Figure 9. The trajectory of torque T_c to T_d in constrained control at $2Hz$

These figures show that the two controllers (saturated and constrained controllers) can provide a stable driving in presence of nonlinear nature of friction, disturbance of the road and actuator saturation. In order to compare the proposed control methods, we will simulate the EPS system with saturated and constrained controllers at $2Hz$ frequency.

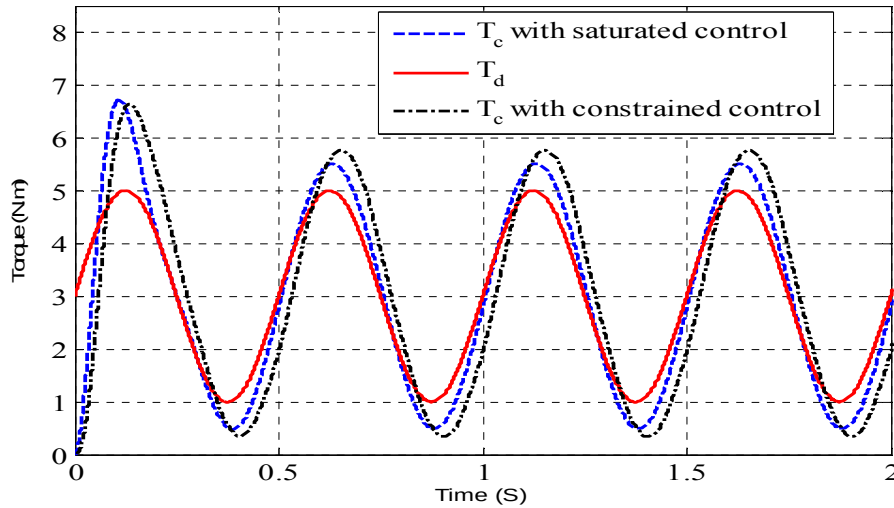


Figure 10. Performances performance of the closed-loop system with saturated and constrained controllers

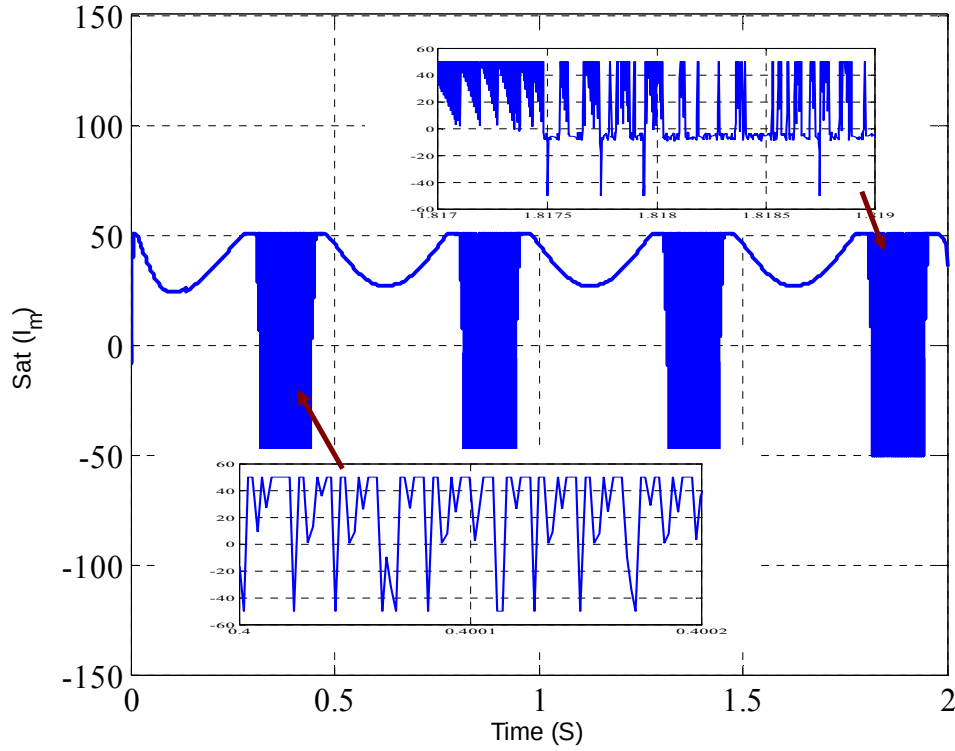


Figure 11. The trajectory of the input control I_m in saturated control case.

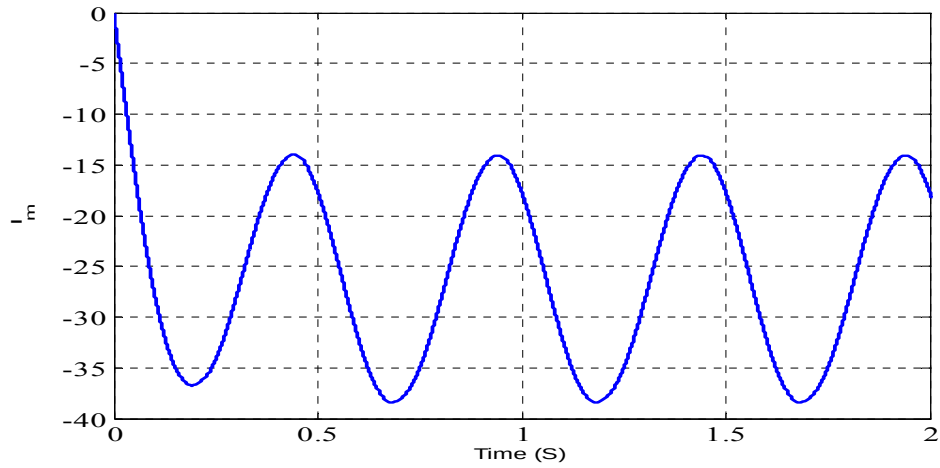


Figure 12. The trajectory of the input control I_m in constrained control case.

These figures show that, despite the presence of actuator saturation, of friction nonlinearity and disturbance of the road, the two approaches (saturated control and constrained control) can give a good steering feeling. The simulation result in Figure 10 illustrates that the

saturated control gives a better and stable driving performance. However, the chattering phenomenon appears for the saturated control (see for example Figure 11 between 0.4 s and 4.5 s), especially actuators with velocity saturation. In this control type, the saturation effect has the sign function form which leads to the chattering that cannot be supported by some actuators. Note that in practical engineering applications, the chattering may appear when sign function is used [28]. Contrary, the constrained control gives a smooth control signal (see Figures 12). Indeed, for actuators protection, the constrained control is recommended.

6. Conclusion

This work has presented a fuzzy control design for EPS systems via state-feedback under actuator saturation and external disturbances. The T-S fuzzy system was first used to describe the EPS system. Then, the controller was designed to guarantee the resulting closed-loop system stability in the presence of actuators saturation. The control saturation design is dealt with by using two different methods; the first is the constrained control input method whereas the second is a saturated control input method. Stabilization conditions for a nonlinear EPS system with both constrained and saturated control input cases are proposed in terms of linear matrix inequalities. Simulation results show that both the saturated and constrained control laws can stabilize the resulting closed-loop EPS system and provide a stable driving in presence of friction nonlinearity, disturbance of the road and actuator saturation. Moreover, the saturated control leads to the appearance of the chattering phenomenon because of the saturation affects. Whereas, the constrained control gives smooth control input signal.

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